FlowDesk — Diagrama de clases

Una sola vista · cada clase aparece una sola vez

CONSULTAS Y ACCESO

Report
- reportId: String
- startDate: Date
- endDate: Date
- reportType: ReportType

+ generateDailyReport(): void
+ generateWeeklyReport(): void
+ generateMonthlyReport(): void

Dashboard
- orders: List<WorkOrder>
- quantityByStatus: Map<OrderStatus, Integer>

+ showSummary(): void
+ filterOrders(status: OrderStatus): List<WorkOrder>
+ showPendingOrders(): void

User
- userId: String
- username: String
- protectedPassword: String

+ login(): boolean
+ logout(): void
+ validateCredentials(): boolean

PAGOS Y CAJA

CashRegister
- movements: List<CashMovement>
- income: double
- expenses: double

+ calculateIncome(): double
+ calculateExpenses(): double
+ calculateBalance(): double

CashMovement
- movementId: String
- type: MovementType
- amount: double
- date: Date
- reason: String
- paymentMethod: PaymentMethod
- payment: Payment [0..1]

+ registerMovement(): void
+ search(id: String): CashMovement

Payment
- paymentId: String
- order: WorkOrder
- amount: double
- date: Date
- paymentMethod: PaymentMethod

+ registerPayment(): void
+ validateAmount(): boolean
+ getPaymentDetails(): String

ORDENES DE TRABAJO

WorkOrder
- orderId: String
- date: Date
- customer: Customer
- details: List<OrderDetail>
- payments: List<Payment>
- total: double
- balance: double
- status: OrderStatus

+ create(): void
+ changeStatus(newStatus: OrderStatus): void
+ calculateTotal(): double
+ calculateBalance(): double
+ search(id: String): WorkOrder

OrderDetail
- detailId: String
- product: Product [0..1]
- service: Service [0..1]
- itemName: String
- quantity: int
- description: String
- unitPrice: double

+ addDetail(): void
+ updateQuantity(quantity: int): void
+ calculateSubtotal(): double

Customer
- customerId: String
- name: String
- phone: String
- idNumber: String
- address: String

+ register(): void
+ search(id: String): Customer
+ update(data: Customer): boolean

CATALOGO

Product
- productId: String
- name: String
- category: Category
- description: String

+ register(): void
+ search(id: String): Product
+ update(data: Product): boolean

Service
- serviceId: String
- name: String
- category: Category
- description: String
- price: double

+ register(): void
+ update(data: Service): boolean
+ search(id: String): Service

Category
- categoryId: String
- name: String
- description: String

+ create(): void
+ updateCategory(): void
+ listProducts(): List<Product>

FECHA Y TIPOS AUXILIARES

«enumeration» **ReportType**
DAILY
WEEKLY
MONTHLY

«enumeration» **OrderStatus**
WAITING
APPROVED
IN_PROGRESS
COMPLETED
DELIVERED

Date
- day: int
- month: int
- year: int

«enumeration» **PaymentMethod**
CASH
TRANSFER

«enumeration» **MovementType**
INCOME
EXPENSE

COMO LEERLO

Todas las flechas discontinuas representan dependencias «use».
La flecha apunta a la clase o tipo utilizado.
− atributo privado + método público [0..1] opcional

Cada detalle selecciona Product O Service.
Date es una clase didáctica; las enumeraciones son listas de opciones.
Propuesta: varios detalles y pagos por orden, pendientes de confirmar.